Lecture 3
Loops

2ip90

Fraud

- collaboration and (re)using other's work is professional practice in software development
- BUT we have an educational situation and have to assess your knowledge and skills, therefore:
  - do not copy work from others (including AI)
  - do not offer your work to others (is considered fraud as well!)
- make an agreement with your partner how you will be using AI in your work

most important:

make sure you are learning

in the exam you have to do it on your own

A Problem

- Computer has a fixed set of elementary instructions
- A program is a (long) sequence of these instructions
- Do we have time to write these programs?



- A computer can perform one instruction every nanosecond (1 GHz). Suppose a dedicated programmer writes one instruction every second. How long would it take him/her to write a program that runs for 1 second?
- 30 years...

Solution? ... Repetition!

Repetition



Repetition (while)

- Aim: execute some statements more than once
- Means: while-construct
- Meaning: repeat loop-body while guard is true; guard is inspected before each execution

Repetition example

Count from 0 up to 20

```
int x = 0;

while (x < 21) {
  System.out.println(x);
  x = x+1;
}

System.out.println(
  "ready!");
```

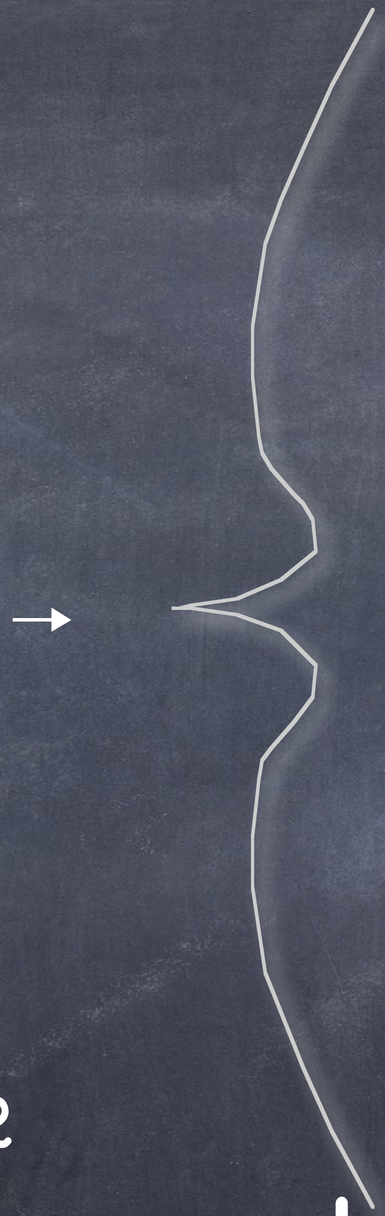
output

```
0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
ready!
```

Repetition example

Count from 0 up to 2

```
int x = 0;
while (x<3) {
  System.out.println(x);
  x = x+1;
}
System.out.println("ready!");
```



```
int x = 0;

(x<3) ? → yes
System.out.println(x);
x = x+1;

(x<3) ? → yes
System.out.println(x);
x = x+1;

(x<3) ? → yes
System.out.println(x);
x = x+1;

(x<3) ? → no

System.out.println("ready!");
```

output

0

1

2

ready!

Repetition - syntax

```
while ( <boolean expression> ) {          "guard"

   <statements>                           "loop body"

}
```

- boolean expression as in if-construct: something with the value true or false
- examples:
  x < 2
  x == y
  0 < x && x < 10

see Coding Style

- on Canvas

Different techniques to control the number of iterations

Fixed number of iterations

```
class Hello3Times {
    // outputs Hello 3 times
    void showHello() {
        int n; // number of Hellos printed

        n = 0; // set guard variable

        while (n < 3) {  //check guard
            System.out.println("Hello");
            n = n + 1; // update guard variable
        }
    }
}
```

output

```
Hello
Hello
Hello
```

User chosen number of iterations

```
import java.util.*;
class HelloNTimes {
    Scanner sc = new Scanner(System.in);

    // outputs Hello times times
    void showHello() {
        int times; // guard limit variable
        int n;    // number of Hellos printed

        System.out.println("Choose number of times");
        times = sc.nextInt(); // choose number of times
        n = 0;      // set guard variable
        while (n < times) {  // guard
            System.out.println("Hello");
            n = n + 1; //update guard variable
        }
    }
}
```

Repeat while user wants to

```
import java.util.*;
class HelloTillStop {
    Scanner sc = new Scanner(System.in);

    // outputs Hello while user wants to
    void showHello() {
        String choice; //guard variable

        System.out.println("For Hello type h, anything else to stop");
        choice = sc.next(); // set guard variable
        while ("h".equals(choice)) {  // check guard
            System.out.println("Hello");
            choice = sc.next(); //update guard variable
        }
    }
}
```

Repeat until result

```
import java.util.*;
public class AccountCompound {
    Scanner sc = new Scanner(System.in);
    // calculate  balance build-up until one million is reached
    void handleBalance() {
        double balance;
        double rate = 5; // interest rate as percentage
        int n;     // number of years passed

        System.out.println("Type amount for balance");
        balance = sc.nextDouble();
        n = 0;
        while (balance < 1000000) {
            System.out.println("Balance after "+n+" years: "+balance);
            n = n + 1;
            balance = balance + (rate * balance)/100;
        }
        System.out.println("Millionaire after "+n+" years");
    }
}
```

Repeat until user says to stop

- not possible (unless reading input from user every iteration of the loop)
- note: the following does not work

```
// does not work!
import java.util.*;
class HelloTillInterrupt {
    Scanner sc = new Scanner(System.in);

    // outputs Hello while user wants to
    void showHello() {
        String choice; //guard variable

        System.out.println("For Hello type h, anything else to stop");
        choice = sc.next(); // set guard variable
        while ("h".equals(choice)) {  // check guard
            System.out.println("Hello");

        }
    }
}
```

sum of numbers 1 to n

- How to calculate the sum of the first n numbers?
  - read n from input
- 1 + 2 + 3 + ... + n-1 + n
- make loop and add 1, next cycle 2, next cylce 3, etc.
- add to what?
- to a variable (e.g., sum): an accumulator
- note: can be done with a formula, without a loop
  nevetheless, the loop program follows the description and it is a simple example of an accumulator loop

accumulator example

- also see demo

```
int n = 10; // last number to be added
int i = 1; // counter
int sum = 0; // the sum (accumulator)
while (i <= n) {
  sum = sum + i;
  i = i + 1;
}
System.out.println("Sum of numbers 1 to %d is %d".formatted(n, sum));
```

- the font I use shows <= as ≤, && as ∧, and more
  this is a so-called ligature font (Liga Fira Code in this case)
  many editors (VS Code included) supports this

accumulator example

- also see demo

```
int n = 10; // last number to be added
int i = 1; // counter
int sum = 0; // the sum (accumulator)
while (i ≤ n) {
  sum = sum + i;
  i = i + 1;
}
System.out.println("Sum of numbers 1 to " + n + " is " + sum);
```

- the font I use shows <= as ≤, && as ∧, etc.
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accumulator example

- also see demo

```
int n = 10; // last number to be added
int i = 1; // counter
int sum = 0; // the sum (accumulator)
while (i ≤ n) {
  sum = sum + i;
  i = i + 1;
}
System.out.println("Sum of numbers 1 to %d is %d".formatted(n, sum));
```

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for-loop

- when you want to do something for each member of a set or range of values
- when you know the number of iterations (number of times you execute the loop-body) on beforehand

often written as i++

```
for (int i = 0; i<100; i=i+1) {
  System.out.println(
     "I should not forget the braces around loop bodies."
  );
}
```

for-loop - syntax

```
for ( <initialization> ; <guard> ; <progress> ) {

    <body>

}
```

- <initialization>: first assignment to counter or other variable(s)
- <progress>: often "incrementing" a counter (i=i+1, j=j-1, k=k+x, etc.)
- <guard>: boolean expression as in while

General structure of a loop

1. initialization: starting values for the variables involved
2. guard: conditional expression determining when to continue or stop
3. body: work that has to be repeated
4. progress: changing variable(s) of the guard to make progress

do-while loop

- when you have to do something at least once

```
double balance;
String choice;

balance = 0;
  do {
    System.out.println("Type amount for deposit.");
    deposit = sc.nextDouble();
    balance = balance + deposit;
    System.out.println("New: " + balance + ".");

   System.out.println(
   "Do you want to make another deposit? Type y/n");
    choice = sc.next());
  } while (choice.equals("y"));
```

do-while loop

- when you have to do something at least once

```
int som = 0;
int cijfer = 0;

do {
    som = som + cijfer;
    System.out.println("type cijfer, -1 om te stoppen");
    cijfer = scanner.nextInt();
} while (cijfer > 0);
```

Remarks

- i++ and i+=1 are shorthand for i=i+1

idea of += works with many operators, such as *=, -=, and %=

- don't mix them! i=i++ is wrong!
- It is in some cases allowed to write the body of a loop or the branch of an if without braces ({ }).
  Don't do this. It easily leads to errors.



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Quiz: power of loops

- Which construct is the most powerful (can express algorithms others can not express)?

A while

B for

C do-while / while

D none, they are all equal in power

Quiz: power of loops

- Which construct is the most powerful (can express algorithms others can not express)?
  - A while 
  - B for 
  - C do-while / while 
  - D none, they are equal in power 

Comparing while and for

```
int i;
…
i = 0;
while ( i < 10 ) {
    System.out.println( i*i );
    i++;
}
```

equivalent ⟷

```
for ( int i = 0; i < 10; i++ ) {
    System.out.println( i*i );
}
```

```
while ( guard ) {
    body
}
```

equivalent ⟷

```
for ( ; guard ; ) {
    body
}
```

Termination

- will loops always terminate?

```
int i = 1;
while (i > 0) {
  System.out.println(i * i);
  i = i + 1;
}
```

it will not terminate because i starts out greater than zero and it will never decrease, so i > 0 keeps being true

actually, this is not true. i will decrease when it reaches the maximal value of an int; then it will become a negative number